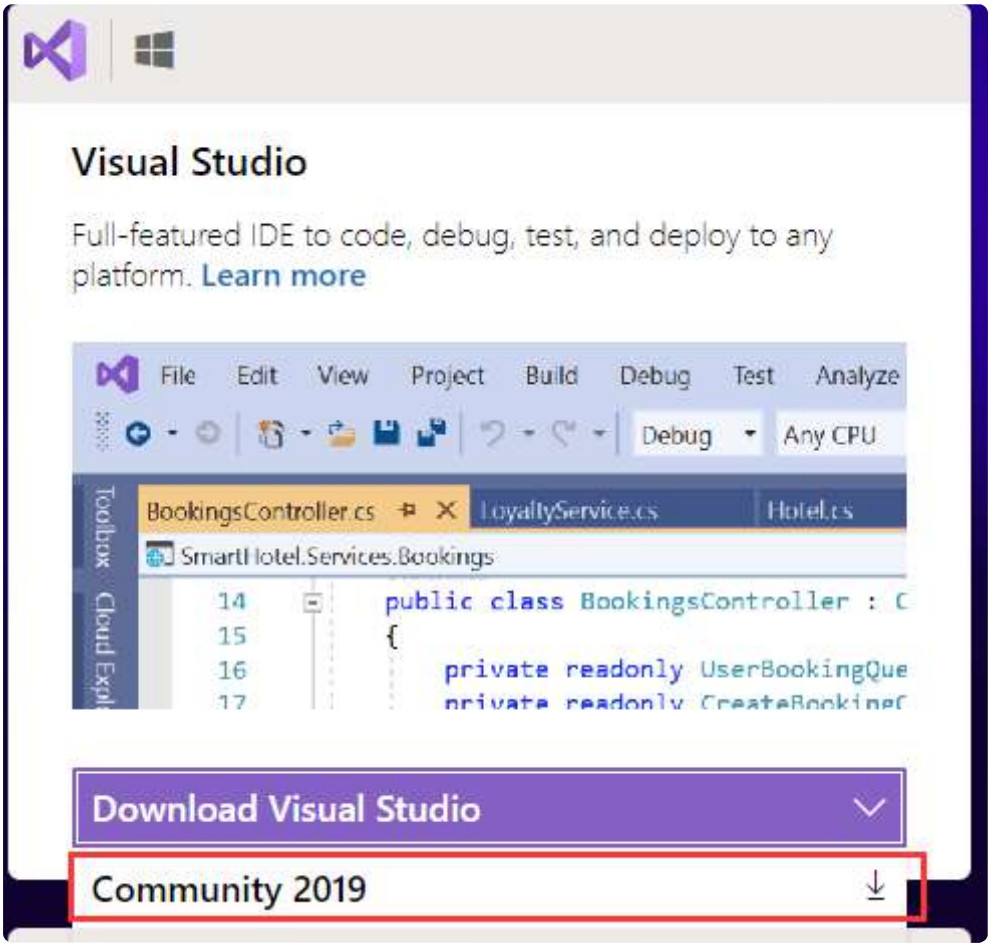
Setup SDL2 with VS2019 and vcpkg

Taowyoo 1 Feb • 3 min read

Setup tools

Install Visual Studio Community 2019

1. Download the installer of Visual Studio from https://visualstudio.microsoft.com/



1. Install Visual Studio with from C++ selected as follow:

0 0 0

Install vcpkg

1. Install Git
    1. Download and run Git installer from https://git-scm.com/downloads
    2. Select all default options during the Git Setup Wizard to install the Git.
2. Clone and build vcpkg to where you like:

```
git clone https://github.com/microsoft/vcpkg
.\vcpkg\bootstrap-vcpkg.bat
```

For an example, I clone it under `C:\dev\`

1. In order to use vcpkg with Visual Studio, run the following command (may require administrator elevation):

```
.\vcpkg\vcpkg integrate install
```

Install SDL2 through vcpkg

In the folder where you cloned the vcpkg:

```
./vcpkg.exe install sdl2:x64-windows
./vcpkg.exe install sdl2-image:x64-windows
./vcpkg.exe install sdl2-ttf:x64-windows
```

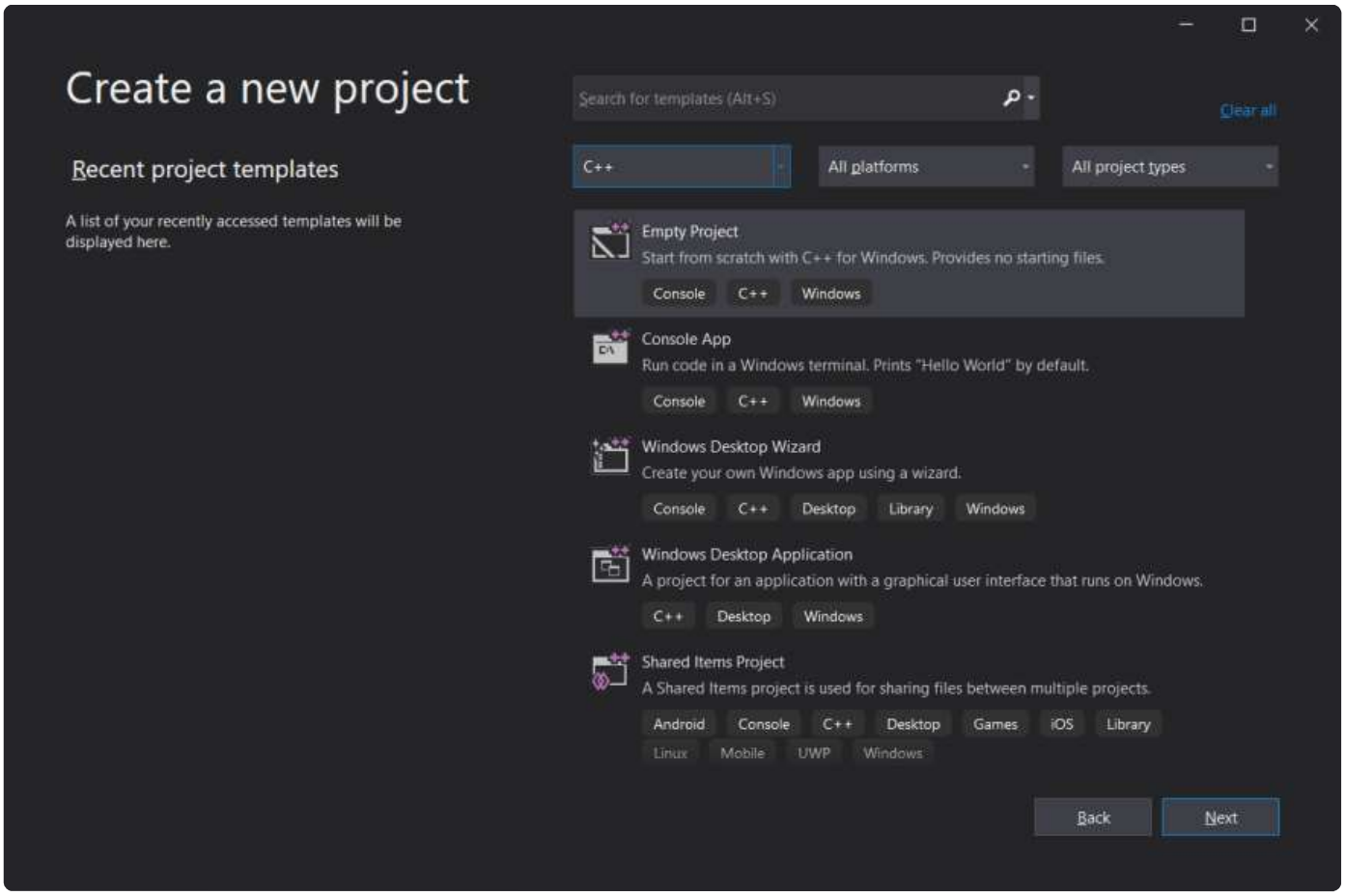
If you want to install the 32-bit version of SDL2:

```
./vcpkg.exe install sdl2:x86-windows
./vcpkg.exe install sdl2-image:x86-windows
./vcpkg.exe install sdl2-ttf:x86-windows
```

Setup an example prject in Visual Studio

1. Start up Visual Studio and create a new empty C++ project

0 0 0



Give your project/solution whatever name you'd like and place it where ever you'd like

2. Now, if your default build setting is Debug x86, you may need to change it:

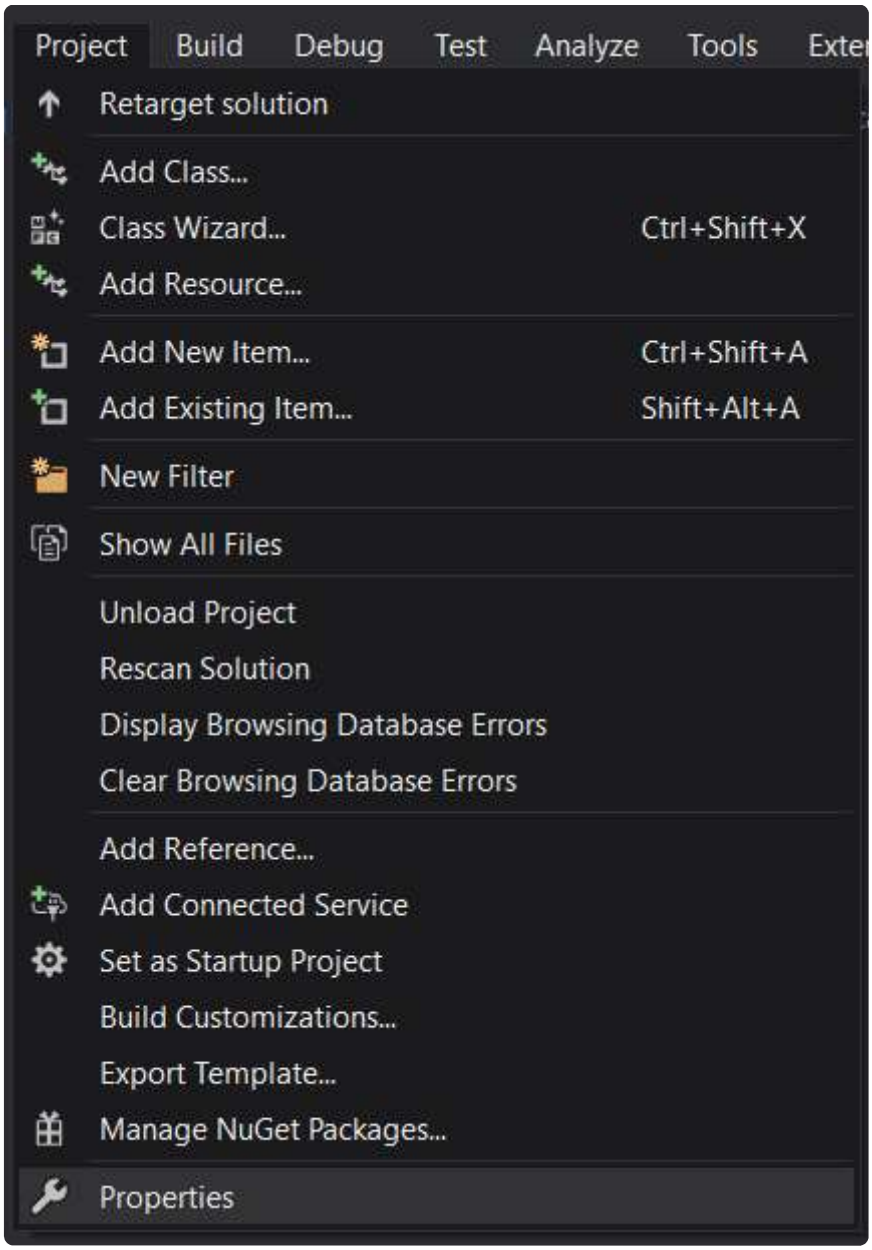
*If you installed the x86 version of SDL2 above, you do not need to do this.

3. Because the [SDL2 redefine the main function](), so the vcpkg won't help us auto link the SDL2main.lib

4. Let's manually link the SDL2main.lib

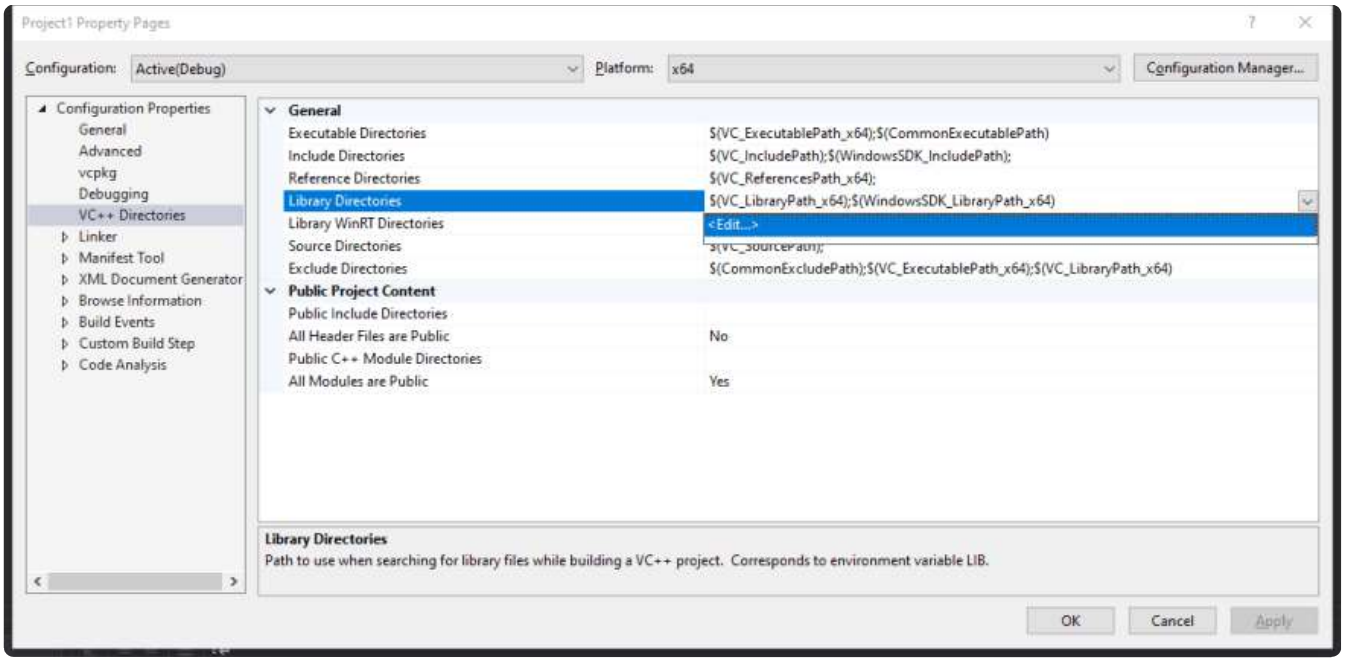
    1. Go to project properties:

0 0 0



2. Go to Configuration Properties -> VC++ Directories -> Library Directories -> Edit.

0 0 0

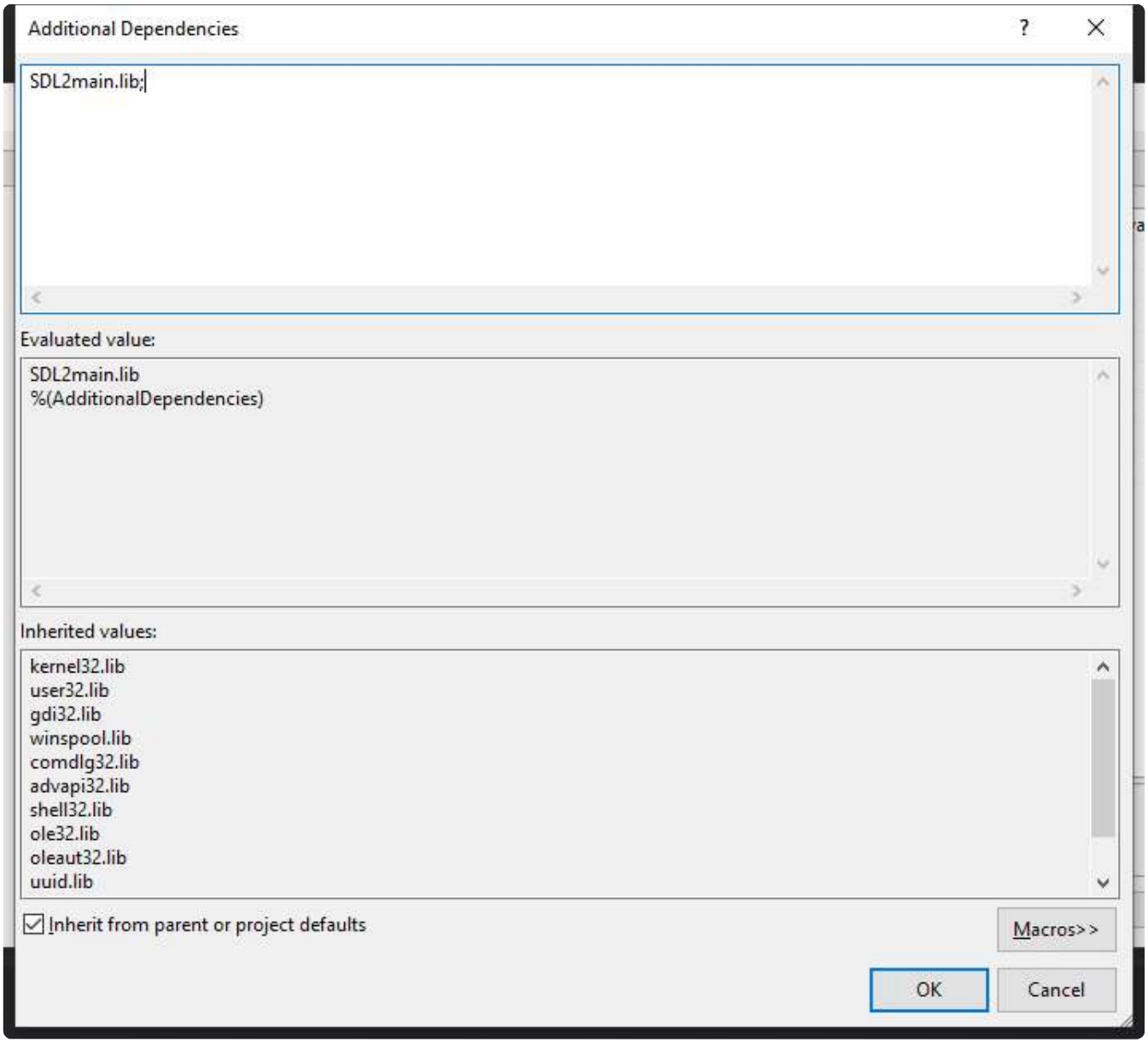


3. And then add the directory where SDL2main.lib located:

   `$(VcpkgCurrentInstalledDir)$(VcpkgConfigSubdir)lib\manual-link\`

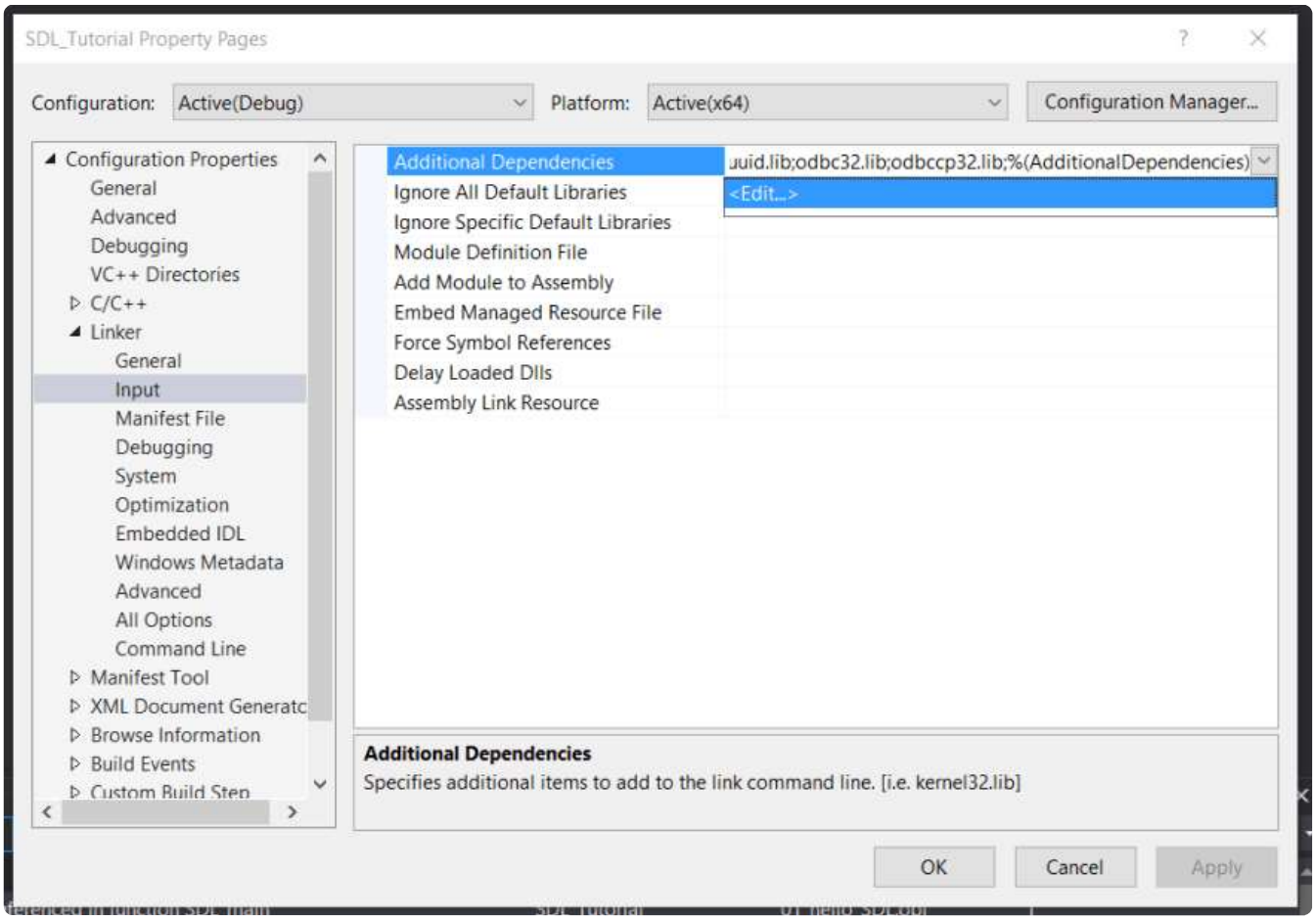
4. Then told linker to link the SDL2main.lib, go to Configuration Properties -> Linker -> Additional Dependencies -> Edit.

0 0 0


Additional Dependencies
SDL2main.lib;
Evaluated value:
SDL2main.lib
%(AdditionalDependencies)
Inherited values:
kernel32.lib
user32.lib
gdi32.lib
winspool.lib
comdlg32.lib
advapi32.lib
shell32.lib
ole32.lib
oleaut32.lib
uuid.lib
Inherit from parent or project defaults
Macros>>
OK
Cancel

Then add

0 0 0

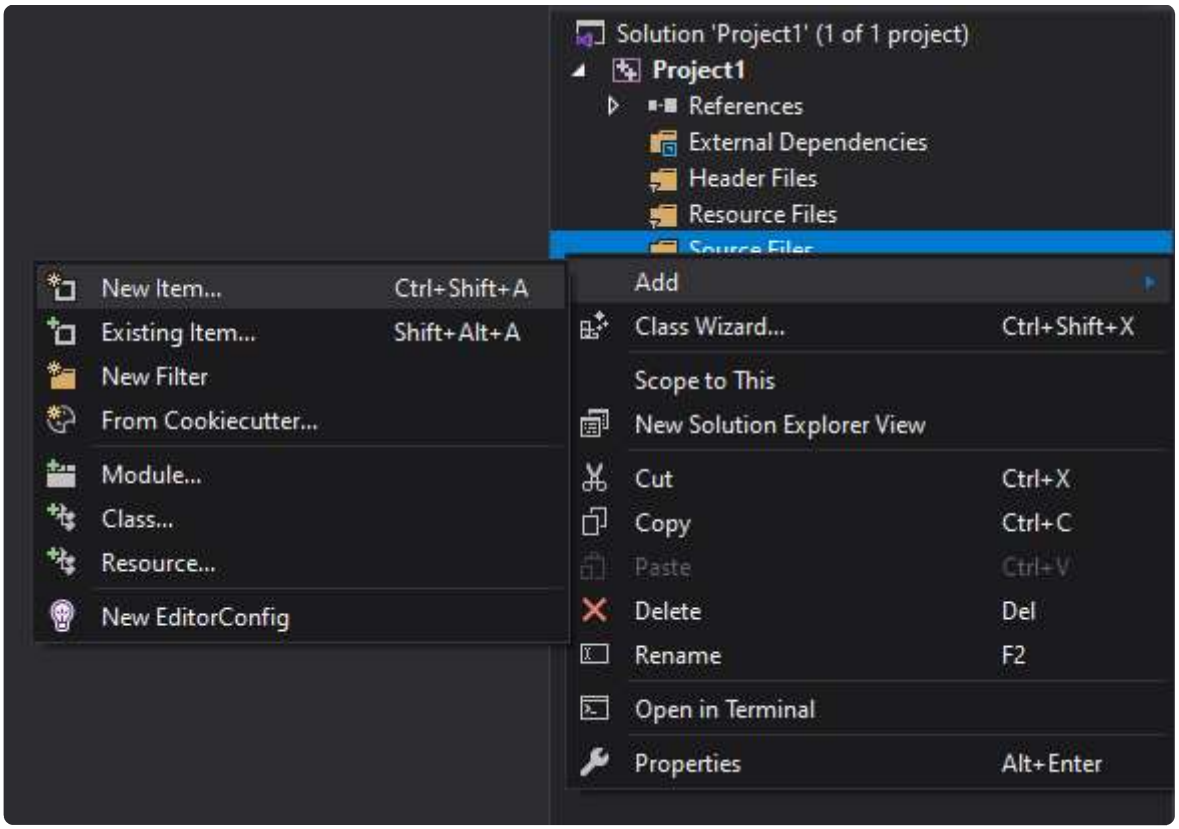


5. Now we finished all we need before coding, close the property page.

5. Finally, create an example program to test the configuration

   Create a new source code file:

0 0 0



Then add following code into it:

```
//Using SDL and standard IO
#include <SDL2/SDL.h>
#include <stdio.h>

//Screen dimension constants
const int SCREEN_WIDTH = 640;
const int SCREEN_HEIGHT = 480;

int main(int argc, char* args[])
{
    //The window we'll be rendering to
    SDL_Window* window = NULL;

    //The surface contained by the window
    SDL_Surface* screenSurface = NULL;

    //Initialize SDL
```

0 0 0

```
        printf("SDL could not initialize! SDL_Error: %s\n", SDL_GetError());
    }
    else
    {
        //Create window
        window = SDL_CreateWindow("SDL Tutorial", SDL_WINDOWPOS_UNDEFINED, SDL_WINDOWPOS_U
        if (window == NULL)
        {
            printf("Window could not be created! SDL_Error: %s\n", SDL_GetError());
        }
        else
        {
            //Get window surface
            screenSurface = SDL_GetWindowSurface(window);

            //Fill the surface white
            SDL_FillRect(screenSurface, NULL, SDL_MapRGB(screenSurface->format, 0xFF, 0xFF

            //Update the surface
            SDL_UpdateWindowSurface(window);

            //Wait two seconds
            SDL_Delay(2000);
        }
    }

    //Destroy window
    SDL_DestroyWindow(window);

    //Quit SDL subsystems
    SDL_Quit();

    return 0;
}
```

After press Ctrl+F5, if all right, you could see a white window pop up for about 2 seconds and then closed.

Discussion (1)

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0 0 0

theundergroundsorcerer • Feb 4

1. I believe that for debug configuration is should be SDL2maind.lib and SDL2main.lib
2. What is the downside of putting #define SDL_MAIN_HANDLED in the header instead of manually linking SDL2main.lib? Does it do any difference?

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Taowyoo

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0 0 0